ME EN 5960/6960-002: Aerospace CFD

Project #1

Due date: Wednesday, October 28th by 11:59pm

Submission

Assignments can only be submitted on Gradescope, which can be accessed through Canvas. Submissions will be automatically locked at the due date given above.

Use of Generative AI

This assignment expects you to utilize generative AI to progress through it. The goal of this AI use is to help you spend more time on important topics while minimizing time spent on less important/menial tasks throughout the assignment. The decision on what is a less important/menial task lies with the instructor and will be clarified in this section and throughout the assignment. You are responsible for verifying that any AI generated code implements the equations and logic you intended. Remember, we are using AI to help you learn. Please do not use it to skip the learning process. Note that sometimes, you will be learning how to best use AI as a tool for your engineering careers.

As a baseline, you may never:

1. Directly upload this assignment anywhere, including as part of an AI prompt.
2. Copy any part of this assignment and include it as part of an AI prompt. Unless explicitly banned as part of the problem, you may write a prompt in your own words based on part of the assignment.

You may always:

1. Use AI to help you understand what a problem is asking or explain some concept that is required for the problem if you follow all other AI rules. For example, if in the assignment you were asked whether a scheme is explicit/implicit, you may prompt AI with "Please explain the difference between an explicit vs. implicit scheme". You may not prompt it with the scheme you are using and ask whether it is explicit or implicit.
2. Use AI to help you understand how a programming language works, but not how to specifically solve your problem in that language. "How do I write a "for" loop in Python" is fine, "Write a CFD solver for me in Python" is not.

Recommended AI

You may use any AI at your disposal, but I recommend using the University of Utah ChatGPT workspace available to all students for free through https://ai.utah.edu/tools/chatgpt/get-started.php. This protects your data and provides a significant amount of free monthly usage. Always monitor your usage limits through settings -> usage, I cannot help you if you run out. At the time of writing, I recommend using ChatGPT 5.6 Sol Medium as it is highly effective but relatively low on resource use. Adjust appropriately as the difficulty of the problem scales.

Programming Language

I *highly* recommend using Python throughout this course, but MATLAB is acceptable if needed.

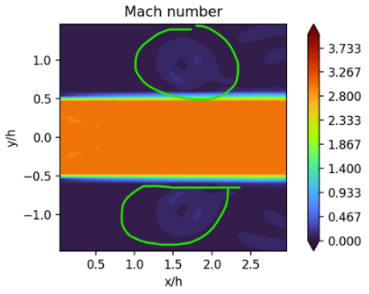
Project: In this project, you are provided with a CFD code that solves the two-dimensional Euler equations similar to what you produced yourself in Assignment #2. You will utilize this code to study compressible flow physics corresponding to a supersonic jet. Note that when run, the code produces output in a directory called "euler_jet_output" which includes two-dimensional contour plots at discrete times, a video of the entire flow, centerline variable plots, and full grid data at the final timestep.

Problem #1: We will first assess the role of boundary and initial conditions on the simulation.

a) Run the code as provided and view the results. You should see a jet exiting from the center of the left boundary and traveling rightwards towards the outlet.

b) In the default case, the pressure ratio between the jet inlet and the ambient air is 1. According to compressible flow physics, what should we expect to see happen in this case? Specifically, comment on how isentropic the flow should be and whether shocks/expansion waves should be present. Do you see the expected physics in your simulation?

c) Study the Mach number contours of your previous results. You should notice the presence of vortices that form around the jet, as illustrated in the image to the right. It is unclear whether the shape and size of these vortices is correct, or if they are an artifact of our simulation. Specifically, we set the upper and lower boundary conditions as Dirichlet boundaries at the ambient condition. However, it may not be accurate to assume ambient conditions when we are this close to physics that does not match the ambient conditions (i.e., the jet). Adjust the variable DOMAIN_LENGTHS_Y and study its effect on the overall flow. Include images labelled with the value of the domain length in Y and identify how it varies the resulting solution. Comment on the role of domain size and boundary conditions on solutions both generally and with respect to this current configuration.



d) Set DOMAIN_LENGTH_Y = 3.0 and PRESSURE_RATIO = 1.2 and run the simulation with all other variables set to their initial values. This initial pressure ratio should correspond to an under-expanded jet. Based on the results of your simulation, are you capturing (qualitatively) the expected behavior in your simulation? Describe what features of your flow influence your answer.

e) Note that your previous simulation doesn't show a full diamond, and that some reverse flow seems to be coming in from the right boundary condition. We may be able to alleviate these issues by expanding our domain rightwards, thereby moving our boundary condition away from the location of interest similar to part c. Adjust DOMAIN_LENGTHS_X and study its effect on the overall flow. Include images labelled with the value of the domain length in Y and identify how it varies the resulting solution. Comment on the role of domain size and boundary conditions on solutions both generally and with respect to this configuration.

f) Set DOMAIN_LENGTHS_Y = 3.0, DOMAIN_LENGTHS_X = 6.0, and PRESSURE_RATIO = 0.8. This initial pressure ratio should correspond to an over-expanded jet. Based on the results of your

simulation, are you capturing (qualitatively) the expected behavior in your simulation? Describe what features of the flow influence your answer.

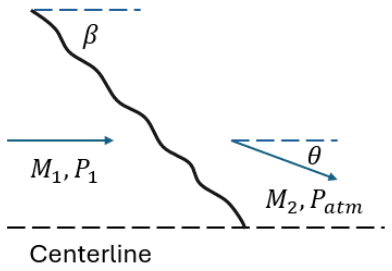
g) Note that your simulation takes some time to reach a quasi-steady state. This isn't an issue for your low-resolution simulation, but if wanted to use more points to get a more accurate solution, this is a lot of wasted time just to reach steady state. Often, we can use a well thought out initial condition to help speed this up and get a more accurate answer faster. Change the variable INITIAL_JET = 1 in the code, which initializes a horizontal strip with height equal to the jet height and flow conditions equal to the jet inlet inside your domain. Estimate how long it takes the simulation to roughly reach steady state for INITIAL_JET = 0 vs. INITIAL_JET = 1. Comment on the role that selecting an appropriate initial solution played and when such an initial solution is most useful. Note not only time to steady-state, but any other resulting changes to the flow.

h) Run the code at $M = [1.5, 3.0, 6.0]$. Estimate the length of a "diamond" from your simulations and plot this length as a function of Mach number. Describe any trends that you see.

i) Run the code at the original Mach number ($M = 3.0$) but vary $\gamma = [1.2, 1.4, 1.6]$. Note that $\gamma = 1.2$ is representative of the value for combustion products, $\gamma = 1.4$ is representative of air, and $\gamma = 1.6$ is representative of noble gases (e.g., Argon). Describe any trends that you see and connect these trends to the types of gases that may be exhausted out of a supersonic jet.

Problem #2: Next, we will try to validate our simulations using compressible flow theory.

a) Focus on the conditions used for the overexpanded jet in problem 1g) (i.e., return to $M = 3.0, \gamma = 1.4$). After exiting from the left boundary, the pressure is too low compared to the atmospheric pressure. Therefore, an oblique shock occurs to increase the pressure up to atmospheric conditions. In the process, the Mach number, flow direction, temperature, and density also change. A simplified image of what happens with this first shockwave is provided on the right. Utilizing the shock relations from Lecture 11, the ratio of jet inlet pressure and atmospheric pressure (note, PRESSURE_RATIO in the code is P_1/P_{atm}), and the inlet jet Mach number (3.0), calculate the expected value of the shock angle (β) in degrees. Assume $\gamma = 1.4$.



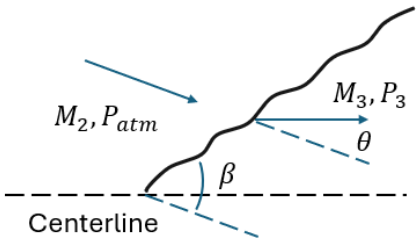
b) Develop a strategy to approximate the angle of the first shockwave from the simulation (remember, you have access to the entire final data field). Do not just estimate it by eye! How does the simulation angle compare to your predicted angle?

c) Using your analytically computed shock angle, calculate the expected static temperature, the static density, and the Mach number after the first shock. Compare these directly to the values from the simulation. Report the calculated values, the simulated values, and the % error for each of these quantities in a table.

d) From the simulation data, estimate the thickness of the first shockwave. You can do this however you'd like but pick a consistent method and justify that method in words. Specifically note how you decide where the shock "starts" and "ends". Note: Any reasonable definition is fine, I just want to see that you've given it some thought. How thick "should" a shockwave be?

e) Perform a grid convergence study. Note, we are not trying to formally prove order of accuracy of the scheme, but rather, are identifying whether our solution converges to the analytical solution as we increase the resolution. You can change the grid resolution by adjusting CELLS_PER_HEIGHT. At a minimum, run cases where it equals 16, 32, 64, but ideally go to higher resolutions as time permits. Plot the post-shock states, shock angle, and shock thicknesses as a function of grid resolution and compare them to your analytical solutions and expectations.

f) Starting from the states you just calculated after the first shock, calculate the states after the second shock, visualized to the right (Mach number, static temperature, static pressure, static density). Note that you no longer know the pressure after the shock, but you do know that the shock must return your flow to being parallel with the centerline, and therefore it must turn it by the same angle that the first shock turned the flow.



g) Compare your results from 2f) with the results from your highest resolution simulation. Comment on how well you are able to predict these quantities. Do the predictions match as closely as they do for the first shock calculation? What could be the cause of this behavior?

Problem #3: This is your opportunity to do something fun with the code. You may pick whatever you find interesting, just don't overdo it (i.e., don't spend 10 hours on this unless you really want to, this effort should be closer to 3-5 hours). For example, you may add the box from Assignment #2 at the center to see how it interacts with the jet. You can convert the right boundary condition to a wall to study rocket exhaust hitting the ground. You can modify the left inlet boundary condition to one that changes over time to see the effect of an increasing Mach number or the effect of a pressure ratio that goes from underexpanded to overexpanded. Perhaps you add a vertical jet to see the interaction of two perpendicular supersonic jets. Whatever you choose to do, you must 1) report what you've implemented and (mathematically) how, 2) show what the resulting flow looks like and identify any interesting features with two-dimensional contours and one-dimensional plots/tables tracking key quantities, 3) attempt to use thermodynamics or fluid mechanics to help describe/predict your flow (equations from outside the course are very welcome), and 4) study how the behavior changes as a function of variables like grid resolution, boundary conditions, etc. Ultimately, I leave the final choice of what you will do up to you, but I am happy to discuss in-person or over email. You should use AI to help understand the current code and understand where to make changes, but those changes should be implemented by you with AI acting only as an assistant.